

# **Phishing URL Detection and Analysis Sandbox**

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**Technologies Used:**

Python • Kali Linux • Requests

**Project Status:**

Implemented & Tested

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## I. Declaration

I declare that this project titled “**Phishing URL Detection and Analysis Sandbox**” is my original work and has been developed for academic and practical learning purposes. All sources of information used in this project have been appropriately acknowledged and referenced.

## II. Acknowledgment

I would like to express my sincere gratitude to everyone who supported and guided me throughout the development of this project. I appreciate the assistance, knowledge, and encouragement that contributed to the successful completion of the **Phishing URL Detection and Analysis Sandbox**.

### III. Dedication

I dedicate this project to my family, mentors, and everyone who has supported and encouraged me throughout my cybersecurity learning journey. Their encouragement and support have motivated me to continue developing my knowledge and practical skills.

## ABSTRACT

The **Phishing URL Detection and Analysis Sandbox** is a Python-based cybersecurity tool designed to identify suspicious characteristics in URLs. The system analyzes factors such as HTTPS usage, IP addresses, suspicious keywords, URL structure, risk indicators, and redirects. It then generates a risk score and classifies the URL as Low, Medium, or High Risk. The project was developed and tested in a Kali Linux environment to provide practical hands-on experience in phishing detection and URL security analysis.

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## CHAPTER 1

### 1. Introduction

Phishing is a common cybersecurity threat where attackers use deceptive links and websites to trick users into revealing sensitive information. Malicious URLs can be difficult to identify through manual inspection alone. This project develops a **Phishing URL Detection and Analysis Sandbox** using Python to examine suspicious URLs and identify common phishing indicators. The system analyzes URL characteristics, assigns a risk score, and provides results that help users understand the potential risk associated with a URL.

### 2. Problem Statement

Phishing URLs continue to pose a significant security risk because they can appear similar to legitimate websites. Users may unknowingly click suspicious links and expose sensitive information. Manual URL inspection is also time-consuming and may fail to identify hidden indicators of phishing. Therefore, there is a need for a simple and practical tool that can automatically analyze URLs and identify suspicious characteristics before users interact with potentially harmful websites.

### 3. Objectives

#### 3.1 Main Objective

To develop a Python-based sandbox that analyzes URLs and identifies characteristics associated with phishing and potentially malicious websites.

#### 3.2 Specific Objectives

- To analyze the structure and characteristics of submitted URLs.
- To identify common phishing indicators such as suspicious keywords and IP-based URLs.
- To analyze HTTPS usage and URL redirects.
- To calculate a risk score based on detected indicators.



- To classify analyzed URLs according to their level of risk.
- To provide practical hands-on experience in phishing detection and cybersecurity analysis.

#### 4. Significance of the Study

This project is significant because it provides a practical approach to identifying suspicious URLs and understanding common phishing indicators. It can help users recognize potentially dangerous links before interacting with them. The project also provides hands-on experience in Python programming, URL analysis, HTTP requests, risk assessment, and cybersecurity testing. Additionally, the sandbox provides a foundation for future development of more advanced phishing detection and threat-analysis systems.

## CHAPTER TWO:

### 2.0 Overview

Phishing is a cybersecurity threat that uses deceptive URLs and websites to mislead users into interacting with potentially harmful resources. A phishing URL may be designed to look trustworthy while containing characteristics that make it different from a legitimate URL. Examining the structure and behavior of a URL can therefore provide useful information when determining whether it contains suspicious indicators.

This chapter reviews the concepts and techniques relevant to the development of the **Phishing URL Detection and Analysis Sandbox**. The review focuses on URL characteristics that can be examined programmatically, including the use of HTTPS, IP addresses, suspicious keywords, URL length, the use of the @ symbol, multiple subdomains, and HTTP redirects.

The chapter also explains the tools used to develop the project, the findings obtained during testing, how the results are analyzed and presented, the gap addressed by the project, and how the developed system was validated and evaluated.

### 2.1 Tools Used for the Study

The development of the Phishing URL Detection and Analysis Sandbox was carried out using a number of tools and technologies. Each tool had a specific role in the development, execution, and testing of the system.

### 2.1.1 Kali Linux

Kali Linux was used as the main operating environment for the project. It provides a suitable environment for cybersecurity development and testing. The project files, Python environment, and analysis program were all executed within Kali Linux.

The project was stored in a dedicated directory:

`~/phishing-url-sandbox`

This directory contains the application and other project folders used during development.

### 2.1.2 Python

Python was used to develop the URL analysis program. The application contains functions responsible for analyzing URL characteristics, calculating the risk score, analyzing redirects, and displaying the results.

Python was selected because it provides built-in functionality for handling strings, regular expressions, URL parsing, and program logic. It also supports external libraries that can be used to perform HTTP requests.

### 2.1.3 Requests Library

The Python requests library was used for redirect analysis.

The system sends an HTTP request to the submitted URL and allows redirects to be followed. The response is then examined to determine:

- Whether the request was successful.
- The number of redirects encountered.
- The URLs involved in the redirect chain.
- The final URL has reached.
- The HTTP status code.
- Any connection error that occurred.

The project also uses a custom User-Agent:

Phishing-URL-Sandbox/1.0

This identifies requests generated by the project during testing.

#### 2.1.4 Regular Expressions

Regular expressions were used to identify whether a hostname follows an IPv4 address pattern.

The system checks whether a URL uses an IP address instead of a normal domain name. An IP-based URL can be considered a suspicious characteristic during phishing URL analysis.

Regular expressions are also used as part of the logic that examines URL characteristics.

#### 2.1.5 Python Virtual Environment

A Python virtual environment was used to isolate the project's Python dependencies from the rest of the system.

The environment is located inside the project directory as:

```
venv
```

The application was executed using the Python interpreter from this environment.

## 2.2 Findings

The findings from the project are based on the URL characteristics implemented in the current analyzer.

### 2.2.1 HTTPS Analysis

The system checks whether a submitted URL uses HTTPS.

HTTPS provides encrypted communication between a client and a web server. In the project, a URL that does not use HTTPS is treated as a suspicious indicator and contributes to the overall risk score.

For example, the test URL:

<http://192.168.1.100/login/verify-account>

uses HTTP rather than HTTPS. The system therefore identified:

**URL does not use HTTPS**

as one of its findings.

### 2.2.2 IP Address Detection

The analyzer checks whether the hostname is an IP address rather than a domain name.

During testing, the URL:

<http://192.168.1.100/login/verify-account>

was identified as using an IP address.

The system reported:

**URL uses an IP address instead of a domain name**

This finding contributes to the risk score.

### 2.2.3 Suspicious Keyword Detection

The system checks the URL for a predefined set of words that can commonly appear in suspicious URLs.

The keywords currently checked include:

- login

- verify
- verification
- secure
- account
- update
- password
- bank
- confirm
- signin

During testing, the URL contained:

login

verify

account

The system detected all three and added them to the findings.

#### 2.2.4 URL Length Analysis

The analyzer checks the length of the submitted URL.

If the URL is longer than 100 characters, the system identifies it as an unusually long URL and adds this as a finding.

This check helps the analyzer identify URLs that contain excessive characters or additional information that may make their structure difficult to inspect manually.

#### 2.2.5 @ Symbol Detection

The analyzer checks whether the submitted URL contains the @ symbol.

If the symbol is present, the system identifies it as a suspicious URL characteristic and increases the risk score.

This provides another structural check when examining potentially deceptive URLs.

### 2.2.6 Hostname and Subdomain Analysis

The system examines the hostname extracted from the submitted URL.

It checks whether the hostname contains multiple dot-separated sections. If the hostname contains three or more dots, the system can identify it as having multiple subdomains.

This contributes to the overall risk score.

### 2.2.7 Risk Score

The analyzer combines the detected indicators to produce a numerical risk score between **0 and 100**.

The current classification is:

- **0–39: LOW**
- **40–69: MEDIUM**
- **70–100: HIGH**

During testing, the simulated URL:

<http://192.168.1.100/login/verify-account>

received a score of:

**65/100 — MEDIUM**

The score was influenced by the lack of HTTPS, use of an IP address, and the presence of the suspicious keywords login, verify, and account.

### 2.2.8 Redirect Analysis

The project also analyzes HTTP redirects associated with a submitted URL.

The system records redirect information such as:

- HTTP status code
- Original URL
- Redirect destination
- Number of redirects
- Final URL

During testing, the simulated IP address could not be reached from the Kali Linux environment.

The system therefore returned:

#### **No route to host**

Instead of terminating the program, the application reported that the redirect analysis had failed and displayed the connection error.

This demonstrated that the system can handle a failed network request and present the error to the user.

## 2.3 Data Analysis and Presentation

The data produced by the analyzer is presented directly through the command-line interface.

After a URL is submitted, the system displays the URL analysis in a structured format. The output contains the hostname, risk score, risk level, detected findings, and redirect analysis.

The output is divided into two major sections:



## URL Analysis

This section displays:

- URL
- Hostname
- Risk score
- Risk level
- Findings

For example, the system can display:

Risk Score: 65/100

Risk Level: MEDIUM

followed by the suspicious indicators detected.

## Redirect Analysis

The second section displays:

- Success status
- Number of redirects
- Final URL
- HTTP status
- Redirect chain
- Error information when the request fails

This format makes the results easy to read and allows the user to understand why a URL received a particular risk classification.

## Research Gap

The project addresses the need for a simple and practical method of analyzing suspicious URLs using clearly understandable indicators.

Many users may not know how to manually examine the different components of a URL or identify characteristics that could indicate suspicious activity. The developed sandbox provides a basic automated approach by checking multiple URL characteristics and presenting the findings in a structured format.

The project therefore focuses on making URL analysis practical and understandable while providing hands-on experience with Python, URL parsing, HTTP requests, and cybersecurity analysis.

The current system is intentionally focused on the URL characteristics and redirect analysis that have been implemented and tested in the project.

## 2.4 Validation and Evaluation

The system was validated by executing the Python application and submitting URLs for analysis.

Testing was performed using:

<https://example.com>

and the simulated suspicious URL:

<http://192.168.1.100/login/verify-account>

The analyzer successfully processed the submitted URLs and produced risk scores and findings.

The suspicious test URL was particularly useful for evaluation because it contained several characteristics that the analyzer was designed to detect. The system identified the use of HTTP, the IP address, and the suspicious keywords to login, verify, and account.

The redirect analysis was also evaluated. Because the IP address used in the test could not be reached from the environment, the system generated a connection error. The application handled the error and displayed it as part of the redirect analysis rather than stopping unexpectedly.

Based on these tests, the current implementation successfully performs the URL analysis functions that have been developed. The results also demonstrate that the analyzer can provide a structured risk assessment and explain the indicators that contributed to the result.

Further development of the project can be documented separately as new features are implemented and tested.

## CHAPTER THREE:

### 3.0 Overview

This chapter describes the methodology used to develop and test the **Phishing URL Detection and Analysis Sandbox**. It explains the approach followed during the development of the project, the flow of the URL analysis process, and the requirements and specifications of the system.

The project was developed using Python in a Kali Linux environment. The methodology focused on creating a simple URL analysis tool that examines specific URL characteristics, identifies suspicious indicators, calculates a risk score, and performs redirect analysis.

### 3.1 Methodology

The project followed a practical and iterative development approach. The system was developed in stages, with each component being implemented and tested before proceeding to the next stage.

#### Step 1: Project Setup

A dedicated project directory was created in Kali Linux:

```
~/phishing-url-sandbox
```

A Python virtual environment was created to provide an isolated environment for the project. The requests library was installed and used for HTTP requests and redirect analysis.

#### Step 2: URL Analysis Development

The first part of the system was designed to examine the structure and characteristics of a submitted URL.

The analyzer checks:

- Whether HTTPS is being used.
- Whether the hostname is an IP address.
- Whether suspicious keywords are present.
- Whether the URL is unusually long.
- Whether the URL contains an @ symbol.
- Whether the hostname contains multiple subdomains.

Each detected indicator contributes to the URL's overall risk score.

#### Step 3: Risk Assessment

After analyzing the URL, the system calculates a risk score from **0 to 100**.

The score is then classified into three risk levels:

| <b>Risk Score</b> | <b>Risk Level</b> |
|-------------------|-------------------|
| 0–39              | LOW               |
| 40–69             | MEDIUM            |
| 70–100            | HIGH              |

This allows the user to quickly understand the overall results of the analysis.

#### Step 4: Redirect Analysis

The system performs an HTTP request using the Python requests library.

Redirects are followed, and information about the redirect chain is collected. The system records the status code, original URL, redirect destination, number of redirects, and final URL.

If the connection cannot be established, the system captures the error and displays it to the user instead of terminating the analysis.

#### Step 5: Result Presentation

After the analysis is completed, the results are displayed through the command-line interface.

The output contains:

- Submitted URL
- Hostname
- Risk score
- Risk level
- Detected findings
- Redirect information
- HTTP status
- Connection errors where applicable

#### Step 6: Testing

The system was tested using URLs with different characteristics.

A legitimate test URL such as:

<https://example.com>

was used to observe the behavior of the analyzer.

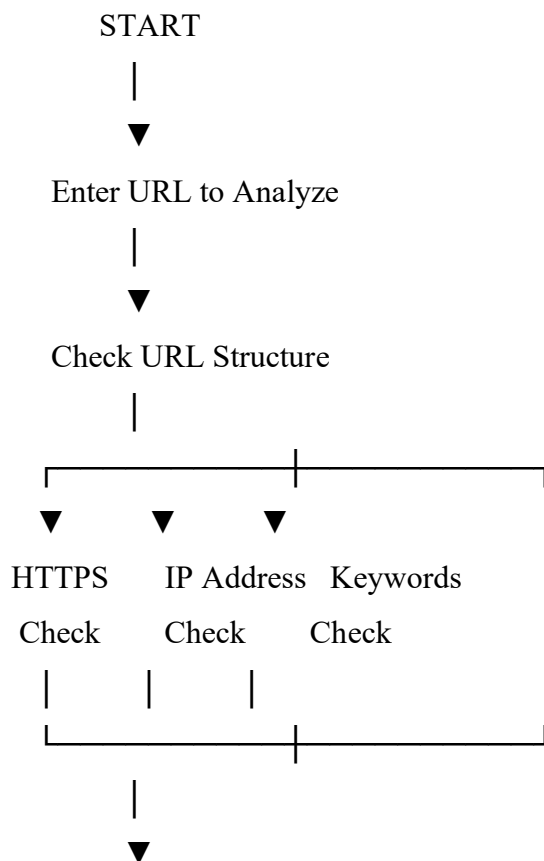
A simulated suspicious URL was also tested:

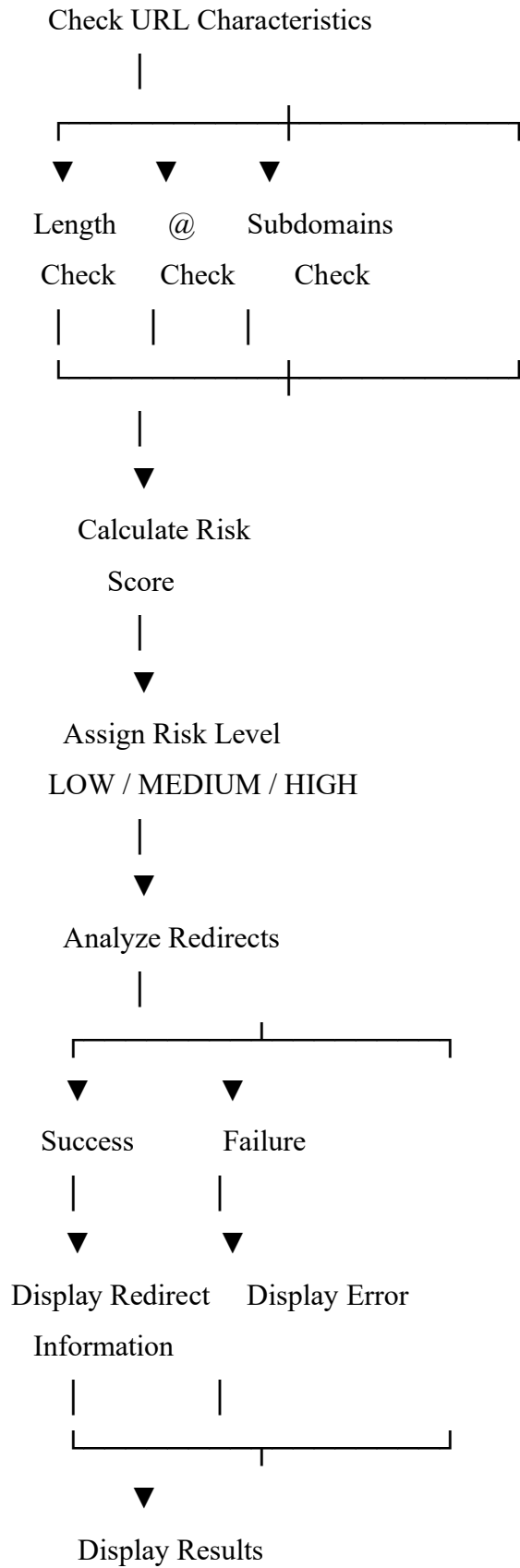
<http://192.168.1.100/login/verify-account>

The system identified several suspicious indicators and assigned the URL a risk score of **65/100 (MEDIUM)**.

### Project Flow Chart

The overall operation of the system can be represented as follows:





|  
▼  
END

## 3.2 Requirements and Specifications

### 3.2.1 Software Requirements

The project requires the following software:

| Software/Technology        | Purpose                             |
|----------------------------|-------------------------------------|
| Kali Linux                 | Development and testing environment |
| Python                     | Development of the URL analyzer     |
| Python Virtual Environment | Isolation of project dependencies   |
| Requests Library           | HTTP requests and redirect analysis |
| Terminal                   | Running and testing the application |

### 3.2.2 Hardware Requirements

The project can be operated on a computer capable of running Kali Linux and Python.

The system requires:

- A computer or laptop.
- Sufficient storage for the project files and Python environment.
- Internet connectivity for URL redirect analysis when testing reachable URLs.
- A working terminal environment.

Project screenshots



```
(venv)-(joy@kali)-[~/phishing-url-sandbox]
$ ls
app  logs  reports  sandbox  screenshots  venv

(venv)-(joy@kali)-[~/phishing-url-sandbox]
$
```

Show your project folders/files.

**Figure 1: Phishing URL Sandbox Project Directory**

```
(venv)-(joy@kali)-[~/phishing-url-sandbox]
$ source venv/bin/activate

(venv)-(joy@kali)-[~/phishing-url-sandbox]
$
```

**Figure 2: Python Virtual Environment**

```

import re
import requests
from urllib.parse import urlparse

# -----
# BASIC URL ANALYSIS
# -----

def analyze_url(url):
    findings = []
    risk_score = 0

    parsed = urlparse(url)

    hostname = parsed.hostname or ""

    # HTTPS check
    if parsed.scheme != "https":
        findings.append("URL does not use HTTPS")
        risk_score += 15

    # IP address check
    ip_pattern = r"^\d{1,3}(\.\d{1,3}){3}$"

    if re.match(ip_pattern, hostname):
        findings.append("URL uses an IP address instead of a domain name")
        risk_score += 25

    # Suspicious keywords
    suspicious_keywords = [
        "login",
        "verify",
        "verification",
        "secure",
        "account",
        "update",
        "password",
        "bank",
        "confirm",
        "signin"
    ]

    for keyword in suspicious_keywords:
        if keyword in url.lower():
            findings.append(

```

**Figure 3. URL Analyzer Source Code**

The screenshot shows the Python code used to analyze URLs. It includes **HTTPS checking**, **IP address detection**, **suspicious keyword detection**, and **risk-score calculation**.

```
(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ ls
app  logs  reports  sandbox  screenshots  venv

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ source venv/bin/activate

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ url_analyzer.py
url_analyzer.py: command not found

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ nano app/url_analyzer.py

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ python app/url_analyzer.py

Phishing URL Sandbox
Enter URL to analyze: "https://www.youtube.com/watch?v=gPhEHbYTMpg"

=====
PHISHING URL ANALYSIS
=====
URL:      http://https://www.youtube.com/watch?v=gPhEHbYTMpg
Hostname: https
Risk Score: 15/100
Risk Level: LOW

Findings:
-----
[!] URL does not use HTTPS

Redirect Analysis:
-----
Success:      False
Redirects:    0
Final URL:    None
HTTP Status:  None

[!] Redirect analysis failed: HTTPConnectionPool(host='x7https', port=80): Max retries exceeded with url: //www.youtube.com/watch?v=gPhEHbYTMpg (Caused by NameResolutionError("HTTPConnection(host='x7https', port=80): Failed to resolve 'x7https' ([Errno -2] Name or service not known)))
=====

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$
```

**Figure 4: Phishing URL Sandbox Execution**

The screenshot shows the sandbox running in Kali Linux and scanning a **legitimate YouTube URL**. The URL was assessed as **low risk with no significant malicious indicators detected**.

```
HTTP Status: None

[!] Redirect analysis failed: HTTPConnectionPool(host='x7https', port=80): Max retries exceeded with url: //www.youtube.com/watch?v=gPhEHbYTMpg (Caused by NameResolutionError("HTTPConnection(host='x7https', port=80): Failed to resolve 'x7https' ([Errno -2] Name or service not known)))
=====

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ http://192.168.1.100/login/verify-account
zsh: no such file or directory: http://192.168.1.100/login/verify-account

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$ python app/url_analyzer.py

Phishing URL Sandbox
Enter URL to analyze: http://192.168.1.100/login/verify-account

=====
PHISHING URL ANALYSIS
=====
URL:      http://192.168.1.100/login/verify-account
Hostname: 192.168.1.100
Risk Score: 65/100
Risk Level: MEDIUM

Findings:
-----
[!] URL does not use HTTPS
[!] URL uses an IP address instead of a domain name
[!] Suspicious keyword detected: login
[!] Suspicious keyword detected: verify
[!] Suspicious keyword detected: account
[!] URL contains multiple subdomains

Redirect Analysis:
-----
Success:      False
Redirects:    0
Final URL:    None
HTTP Status:  None

[!] Redirect analysis failed: HTTPConnectionPool(host='192.168.1.100', port=80): Max retries exceeded with url: /login/verify-account (Caused by NewConnectionError("HTTPConnection(host='192.168.1.100', port=80): Failed to establish a new connection: [Errno 113] No route to host"))
=====

(venv)-(joy@kali)-[~/phishing-url-sandbox]
└─$
```

**Figure 5: Analysis of an Unsafe URL**

The screenshot shows the sandbox analyzing a **simulated unsafe URL** containing an IP address and suspicious keywords. The system detected multiple risk indicators and assigned the URL a **medium-risk score of 65/100**.

## Conclusion

The **Phishing URL Detection and Analysis Sandbox** successfully achieved its main objective of analyzing URLs for common phishing indicators. The system was able to examine URL characteristics, identify suspicious elements, calculate a risk score, classify the URL, and perform redirect analysis. Testing with both a legitimate YouTube URL and a simulated unsafe URL demonstrated that the system can distinguish between URLs with different levels of risk. The project also provided practical experience in Python programming, URL analysis, HTTP requests, and cybersecurity testing.

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